

Problem Set 7 — FBDs & Newton's Laws

Physics 2026–27 • due at the start of class, Tuesday, October 27

NAME _____

DATE _____

How to work this set. Every force problem starts with a free-body diagram: isolate the object as a dot, draw every force acting *on* it (nothing it exerts on anything else), label each arrow with its symbol. Then KDEA with $\Sigma F = ma$ — or $\Sigma F = 0$ if the velocity isn't changing. Force unit: $N = \text{kg} \cdot \text{m}/\text{s}^2$, exactly as Unit 0 promised.

1 Three FBDs

DIAGRAMS

Draw a labeled free-body diagram for each. Arrow lengths should reflect relative magnitudes.

- A textbook resting on a level table.
- A shop sign hanging motionless from a single vertical rope.
- A shopping cart rolling at *constant velocity* across a flat, frictionless floor — nobody touching it. (Careful: how many forces does it actually have?)

DIAGRAMS

2 Equilibrium or not?

FIRST LAW

For each object, state whether $\Sigma F = 0$, and how you know:

- A parked bus.
- A hockey puck sliding across frictionless ice at a constant 4 m/s.
- A Metrorail train pulling away from the station, speeding up.
- One sentence: what belief about moving objects does (b) disprove?

3 Straight $F = ma$

SECOND LAW

- a. A 2.5 kg lab cart feels a net force of 10 N . Find its acceleration.
- b. A 1400 kg car accelerates at 2.5 m/s^2 . What net force is required?

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	$\Sigma F = MA$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

4 Tug of war

NET FORCE

A 6.0 kg crate sits on a frictionless floor. One student pushes with 25 N to the right, another with 10 N to the left. Draw the FBD, then run KDEA for the acceleration — magnitude *and* direction.

K	EVERY GIVEN, WITH UNITS · PICK A POSITIVE DIRECTION
D	THE TARGET & ITS UNIT
E	ΣF WITH SIGNS, THEN $= MA$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

5 Forces at a right angle

VECTORS RETURN

A 12 kg crate sits on frictionless ice. One person pushes east with 50 N ; another pushes north with 120 N at the same time. Workshop III trig, force edition:

- a. Find the magnitude of the net force.
- b. Find its direction (angle north of east).
- c. Find the crate's acceleration — magnitude and direction.

K	EVERY GIVEN, WITH UNITS
D	THREE TARGETS, THREE UNITS
E	PYTHAGOREAN + $\tan^{-1}$, THEN $\Sigma F = MA$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

6 The normal force is not a constant

CONCEPT

A 1.2 kg book rests on a table.

- Find N . What equation actually justifies it — and what condition makes that equation true?
- Now you press straight down on the book with an extra 5.0 N . What is N now?
- One sentence: why is " $N = mg$ " a special case rather than a law?

WORK

Graded on the homework 4-3-2-1 scale. An FBD with a phantom "motion force" on it is the week's classic error — constant velocity needs no force at all.